Forecast-Calibrated Charger-Based Model Predictive Control for Unidirectional Workplace EV Charging

Exact aggregation, continuous demand-charge accounting, and a three-month out-of-sample pilot

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Abstract

This paper studies whether forecast uncertainty can create measurable economic value in charger-based model predictive control (MPC) for unidirectional workplace electric-vehicle charging. The setting intentionally excludes vehicle-to-grid export, battery storage, photovoltaics, and demand response; therefore, economic value means avoided charging cost rather than market revenue. We first derive a sparse charger-level linear program whose cumulative-energy envelopes are an exact projection of the session-level problem when sessions do not overlap on a physical port. The controller carries executed noncoincident and on-peak demand states continuously through each billing month, preventing the repeated daily demand-charge approximation that can severely distort results. A causal four-week same-weekday median forecast is then corrected with one-sided split-conformal residuals for late arrival, early departure, and underpredicted energy. We evaluate six fixed UC San Diego ports over 92 unseen days in July–September 2023. All 552 day–method runs serve the required energy without fallback. The conformal controller costs \$4,177.90, a 28.23% reduction from immediate charging and a 1.47% reduction from point-forecast MPC; its regret to perfect information is 4.43%. Empirical one-sided coverages are 91.5%, 91.5%, and 90.2% against a 90% target. However, the conformal method improves only two of three billing months, so the result is a reproducible pilot rather than evidence of statistical generalization. Synthetic tests preserve the EV-level objective to 1.14×10^{-12} and give a median 2.07× speedup at eight sessions per charger.

Keywords: electric-vehicle charging; charger aggregation; model predictive control; conformal prediction; demand charges; forecast value

1 Introduction

Workplace EV charging is naturally a receding-horizon control problem: sessions arrive throughout the day, their flexibility differs, and monthly demand charges couple otherwise independent days. Session-level optimization is physically transparent but expands with the number of vehicles. Charger-level optimization is smaller, but a daily-energy-only aggregation can move energy across a vehicle's arrival or deadline and consequently report false savings. The first question in this work is therefore structural: under which operating conditions is a charger representation both smaller and exactly feasible?

The second question concerns uncertainty. Classical charging formulations optimize known requests [2, 7], whereas closed-loop station control must act before later sessions are observed. Stochastic EV-station models explicitly consider uncertain arrival, departure, and energy [3]. Recent charging-hub MPC combines machine-learning forecasts with conformal uncertainty sets and reports a roughly 1% improvement over deterministic point forecasts in a richer PV–battery setting [1]. Here we isolate the charger-control effect: there is no storage, PV, export, V2G, or demand-response signal.

Our contributions are:

1. an exact sparse charger projection for nonoverlapping physical-port sessions, together with an explicit disaggregation argument;
2. a rolling MPC implementation that carries executed billing-peak states across days and evaluates only incremental monthly demand cost;
3. a leakage-controlled comparison of no forecast, persistence, historical-median, one-sided conformal, and perfect-information forecasts; and
4. numerical invariance, feasibility, coverage, and scalability audits that make the boundary of the current evidence explicit.

2 Physical and tariff model

2.1 Sets, data, and assumptions

One day is divided into $\mathcal{T} = \{1, \ldots, T\}$ intervals of length $\Delta t = 0.25$ h; $T = 96$. Let $\mathcal{C}$ be the physical chargers and $\mathcal{I}_c$ the sessions assigned to charger c. Session i has arrival a_i, inclusive departure d_i, requested grid energy E_i, and maximum power $\bar{p}_i$. Its availability set is

$$\mathcal{T}_i = \{t \in \mathcal{T} : a_i \le t \le d_i\}.$$

Charging is unidirectional:

$$p_{i,t} \ge 0.$$

There is no export, stationary battery, PV, feeder-network constraint, or demand-response payment. Charging efficiency is already absorbed into the metered energy request. Once a session arrives, $d_i, E_i, \bar{p}_i$ are assumed observable. Most importantly, sessions on each selected physical port do not overlap:

$$i, j \in \mathcal{I}_c,\ a_i < a_j \quad \Longrightarrow \quad d_i < a_j. \tag{1}$$

Equation (1) is checked in data and before every optimization.

2.2 Session-level reference problem

The exact EV-level feasible set is

$$0 \le p_{i,t} \le \bar{p}_i \mathbf{1}_{\{t \in \mathcal{T}_i\}}, \qquad i \in \mathcal{I},\ t \in \mathcal{T}, \tag{2a}$$

$$\Delta t \sum_{t \in \mathcal{T}_i} p_{i,t} = E_i, \qquad i \in \mathcal{I}. \tag{2b}$$

The total site charging power is

$$P_t = \sum_{i \in \mathcal{I}} p_{i,t}. \tag{3}$$

This reference formulation is a linear program (LP) and is used only for invariance and scalability comparisons, not for the rolling paper experiment.

2.3 Exact charger-level cumulative envelopes

Define cumulative delivered energy through interval t for session i. The greatest energy that could have been delivered is

$$U_{i,t} = \min\left\{E_i,\ \bar{p}_i \Delta t\ [\min(t, d_i) - a_i + 1]_+\right\}, \tag{4}$$

and the least energy that must already have been delivered, leaving enough future capacity, is

$$L_{i,t} = \max\left\{0,\ E_i - \bar{p}_i \Delta t\ [d_i - \max(t+1, a_i) + 1]_+\right\}. \tag{5}$$

Aggregate envelopes are $U_{c,t} = \sum_{i\in\mathcal{I}_c} U_{i,t}$ and $L_{c,t} = \sum_{i\in\mathcal{I}_c} L_{i,t}$. With charger power $p_{c,t}$, introduce the sparse cumulative state

$$
\begin{aligned}
&e_{c,0} = 0, && && \text{(6a)}\\
&e_{c,t} = e_{c,t-1} + \Delta t\, p_{c,t}, && c \in \mathcal{C},\ t \in \mathcal{T}, && \text{(6b)}\\
&L_{c,t} \le e_{c,t} \le U_{c,t}, && c \in \mathcal{C},\ t \in \mathcal{T}, && \text{(6c)}\\
&0 \le p_{c,t} \le \sum_{i\in\mathcal{I}_c} \bar{p}_i \mathbf{1}_{\{t\in\mathcal{T}_i\}}. && && \text{(6d)}
\end{aligned}
$$

Under (1), at most one term in the right side of (6d) is nonzero.

Proposition 1 (Exact charger projection)**.** *If sessions on every physical port satisfy* (1)*, the set of aggregate trajectories satisfying* (6) *is exactly the projection of the session-level feasible set* (2) *under* $p_{c,t} = \sum_{i\in\mathcal{I}_c} p_{i,t}$*.*

Proof. For any feasible session dispatch, summing its cumulative energy establishes (6b)–(6c); availability establishes (6d). Conversely, nonoverlap means every occupied charger–time pair has one unique active session. Assign $p_{c,t}$ to that session and zero to all others. The upper envelope prevents premature energy, the lower envelope forces each completed session to have received E_i, and the pointwise bound enforces $\bar{p}_i$. At $t = d_i$, both session envelopes equal E_i; because later sessions have not arrived, their cumulative contribution is zero. Hence each session energy equality holds and the assignment is a feasible disaggregation. □

The recurrence (6b) has $O(|\mathcal{C}|T)$ nonzeros. A dense prefix-sum encoding has $O(|\mathcal{C}|T^2)$ nonzeros even though it represents the same state.

2.4 Continuous monthly tariff state

Let λ_t be the time-of-use energy price and $\mathcal{T}^{\text{on}} = \{65, \dots, 84\}$ the 16:00–21:00 on-peak intervals. The implemented summer tariff uses $\lambda_t = \$0.12628/\text{kWh}$ on peak and \$0.10679/kWh otherwise, noncoincident demand rate $\kappa^N = \$24.48/\text{kW}$, and on-peak demand rate $\kappa^P = \$28.92/\text{kW}$. Additional tariff coefficients are a fraction $\rho = 0.0578$ and volumetric charge $\eta = \$0.00668/\text{kWh}$.

For billing month m, let $z^N_{m,d-1}$ and $z^P_{m,d-1}$ be the *executed* peaks before day d. Their recursion is

$$
z^N_{m,d} = \max\left\{ z^N_{m,d-1}, \max_{t\in\mathcal{T}} P_{d,t} \right\}, \tag{7a}
$$

$$
z^P_{m,d} = \max\left\{ z^P_{m,d-1}, \max_{t\in\mathcal{T}^{\text{on}}} P_{d,t} \right\}. \tag{7b}
$$

The day-d incremental bill contribution optimized and evaluated is

$$
\begin{aligned}
J_{m,d} = (1+\rho)\Delta t \sum_{t\in\mathcal{T}} \lambda_t P_{d,t}&\\
+ (1+\rho)\kappa^N (z^N_{m,d} - z^N_{m,d-1})&\\
+ (1+\rho)\kappa^P (z^P_{m,d} - z^P_{m,d-1})&\\
+ \eta \Delta t \sum_{t\in\mathcal{T}} P_{d,t}.&
\end{aligned} \tag{8}
$$

Thus demand increments telescope:

$$
\sum_{d\in m} \kappa^N (z^N_{m,d} - z^N_{m,d-1}) = \kappa^N z^N_{m,D_m},
$$

and similarly on peak. Treating each day as a fresh bill would instead charge the demand maximum repeatedly and is not used in reported results.

3 Forecast-calibrated rolling MPC

3.1 Causal point forecast

For target day d, the historical-median model uses only the same weekday in the preceding $K = 4$ weeks. For each charger c, it predicts the number of sessions by the rounded median of the four counts. Sessions are ordered by arrival on each historical day. For ordinal j, the point forecast is

$$(\widehat{a}_{cjd}, \widehat{d}_{cjd}, \widehat{E}_{cjd}, \widehat{\bar{p}}_{cjd}) = \mathrm{median}_{\ell=1,\ldots,K}(a_{cj,d-7\ell}, d_{cj,d-7\ell}, E_{cj,d-7\ell}, \bar{p}_{cj,d-7\ell}), \tag{9}$$

using the available ordinal samples. Windows are clipped to the day and sequentially reconciled to retain port nonoverlap.

3.2 One-sided split-conformal correction

We calibrate failure-relevant residuals on ordinal-matched sessions:

$$r_k^a = a_k - \widehat{a}_k, \qquad r_k^d = \widehat{d}_k - d_k, \qquad r_k^E = E_k - \widehat{E}_k. \tag{10}$$

Positive values respectively mean later arrival, earlier departure, or larger energy than forecast. For a residual sample $r_1, \ldots, r_n$, the finite-sample one-sided split-conformal quantile is

$$q_{1-\alpha}(r) = r_{(\,k\,)}, \qquad k = \min\{n, \lceil (n+1)(1-\alpha) \rceil\}\,, \tag{11}$$

where $r_{(k)}$ is the k-th order statistic [5, 6]. The robust session passed to the deterministic MPC is

$$\widetilde{a}_i = \min\{T, \widehat{a}_i + \max(0, q^a)\}, \tag{12a}$$
$$\widetilde{d}_i = \max\{\widetilde{a}_i, \widehat{d}_i - \max(0, q^d)\}, \tag{12b}$$
$$\widetilde{E}_i = \min\{\widehat{E}_i + \max(0, q^E), \bar{p}_i \Delta t(\widetilde{d}_i - \widetilde{a}_i + 1)\}. \tag{12c}$$

This is a conservative forecast transformation, not a joint chance-constrained program. Marginal split-conformal coverage is expected under exchangeability; the chronological experiment reports empirical coverage rather than asserting that workplace sessions are exchangeable. Count under-prediction is measured but no fictitious session is synthesized because its physical window and energy would be unknown.

3.3 Receding-horizon information pattern

At interval t, the work set contains (i) actual connected sessions rebased to their remaining energy and (ii) forecasts whose predicted arrival is later than t. Arrived truth supersedes any overlapping forecast on the same charger. The controller solves the charger LP with the current monthly peak states, executes only the first action, updates delivered energy and (7), and repeats. Planned power is never delivered unless an actual EV is connected. There is no post-hoc replacement of missed energy. This follows the standard receding-horizon structure [4].

LPs can have many economically identical solutions. For reproducible EV/charger comparisons, every solve is lexicographic. First,

$$J^\star = \min_{x \in \mathcal{X}} J(x).$$

Then the controller chooses

$$x^\star_{\mathrm{lex}} \in \arg\min_{x \in \mathcal{X}} \sum_{c,t} \frac{t}{T} p_{c,t} \quad \text{s.t.} \quad J(x) \le J^\star + \epsilon_{\mathrm{lex}}, \tag{13}$$

where $\epsilon_{\mathrm{lex}} = \max(10^{-6}, 10^{-7}|J^\star|)$ USD. The reported objective is always the primary tariff cost.

4 Experimental protocol

4.1 Data and leakage control

The local UCSD charging table contains 340,060 sessions from July 20, 2016 to September 16, 2024, covering 291 ports. The paper pilot uses six fixed ports and 808 sessions during July 1–September 30, 2023. Ports are ranked by activity and energy using May–June only; a data-quality screen additionally requires nonoverlap throughout the loaded evaluation window. This screen uses test-period timestamps only to verify the structural assumption and does not use test costs, energy magnitudes for ranking, or controller outcomes.

Hyperparameter selection is chronological. Candidate $\alpha \in \{0.1, 0.2, 0.3, 0.4\}$ values use April–May residual calibration and June closed-loop validation. After selecting $\alpha = 0.1$, the corrections are refit on the immediately preceding May–June window, with the Q3 test untouched. The final 680 matched residuals give

$$q^a = 17 \text{ slots}, \qquad q^d = 18 \text{ slots}, \qquad q^E = 13.6035 \text{ kWh}.$$

The 90th percentile count-underprediction residual is one session, but is reported rather than converted to a synthetic request.

4.2 Methods and metrics

Six controllers share the same tariff, physical model, state update, and solver:

- **V0G**: immediate maximum-rate charging;
- **Perfect**: future actual sessions supplied as an unattainable information upper benchmark;
- **No forecast**: MPC knows a request only after it arrives;
- **Persistence**: sessions from the same weekday one week earlier;
- **Historical median**: the causal point forecast in (9); and
- **Conformal**: (12) applied to the historical median.

Cost savings and perfect-information regret are

$$S_m = 100\frac{J_m^{\mathrm{V0G}} - J_m}{J_m^{\mathrm{V0G}}}, \qquad R_m = 100\frac{J_m - J_m^{\mathrm{Perfect}}}{J_m^{\mathrm{Perfect}}}. \tag{14}$$

Forecast errors are computed on ordinal-matched sessions; count mean absolute error (MAE) is averaged per charger-day. A run is accepted only if unserved energy is below 10^{-5} kWh, the solver reports optimality at every call, fallback count is zero, and peak states are monotone within each month.

5 Results

5.1 Control value and forecast calibration

Table 1 shows that all MPC variants improve on V0G. The conformal controller reduces cost by 28.23% relative to V0G, closes most of the gap to perfect information, and costs 1.47% less than historical-median MPC:

$$100\frac{4240.03 - 4177.90}{4240.03} = 1.465\%.$$

Its monthly cost improvement over historical median is \$50.32 in July and \$49.87 in August, but it is \$38.06 worse in September. Three monthly pairs are insufficient for a meaningful significance claim.

Table 1: Out-of-sample control results, July–September 2023. "Cost" is the sum of continuous monthly tariff contributions, not a sum of isolated daily demand bills.

Method	Q3 cost (USD)	Saving vs. V0G	Regret vs. perfect
V0G	5821.16	0.00%	45.50%
Perfect	4000.79	31.27%	0.00%
No forecast	4243.88	27.10%	6.08%
Persistence	4317.58	25.83%	7.92%
Hist. median	4240.03	27.16%	5.98%
Conformal	4177.90	28.23%	4.43%

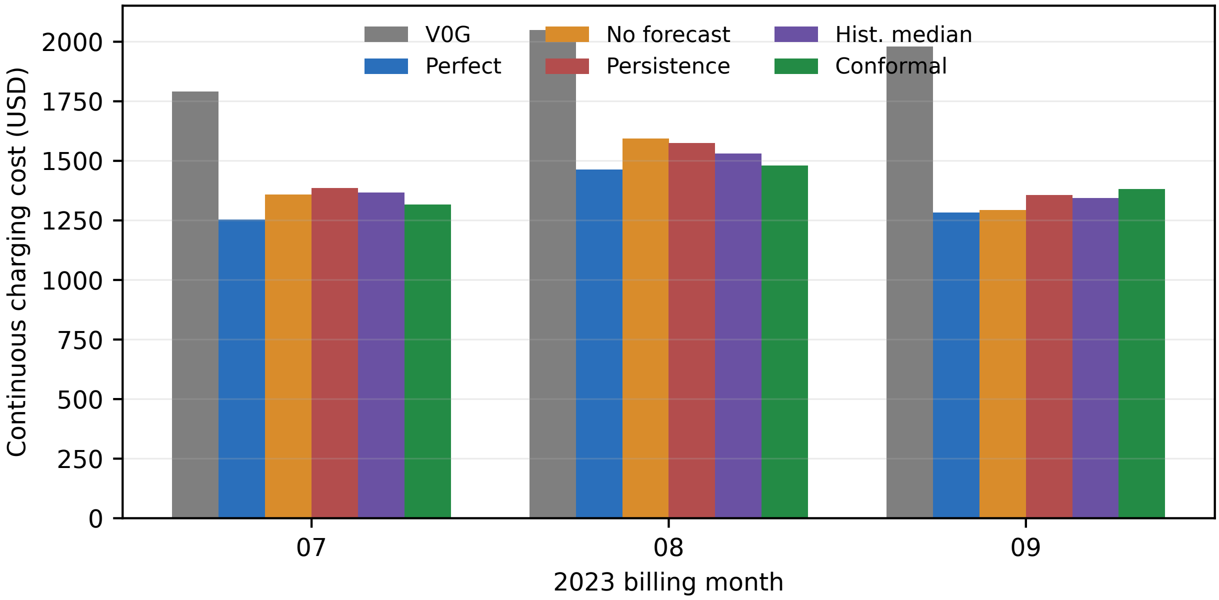

Figure 1: Continuous monthly charging cost. Each method's executed peak state is reset only at the billing-month boundary.

Historical median improves every point metric over persistence (Table 2). The robust transformation increases point MAE, which is expected because its purpose is one-sided risk protection rather than conditional-median accuracy. Against the historical-median point forecast, empirical one-sided coverage is 91.48% for late arrival, 91.48% for early departure, and 90.16% for energy underprediction, close to or above the 90% target.

The validation ranking in Table 3 selects the most conservative candidate. This does not imply that lower α must always be cheaper: shrinking windows and increasing energy can also create excessive early charging and higher peaks.

5.2 Exactness and computational scaling

On 28 real-day comparisons, EV- and charger-level LPs have maximum absolute objective difference 1.14×10^{-12} USD and maximum aggregate-load difference 7.64×10^{-11} kW after the common lexicographic tie-break. Every charger trajectory disaggregates to sessions with exact requested energy.

The charger formulation is not uniformly faster. It is slower at two sessions per port, approximately 1.18× faster at four, and 2.07× faster at eight. The relevant benefit is therefore density-dependent, not an unqualified aggregation speedup.

Table 2: Q3 session forecast diagnostics. Arrival and departure MAE are in 15-minute slots. Robust inputs are intentionally not point-forecast-optimal.

Method	Count MAE (sessions)	Arrival MAE (slots)	Departure MAE (slots)	Energy MAE (kWh)
Persistence	0.962	10.783	11.744	8.031
Hist. median	0.868	9.366	10.687	6.811
Conformal	0.868	18.300	12.203	9.905

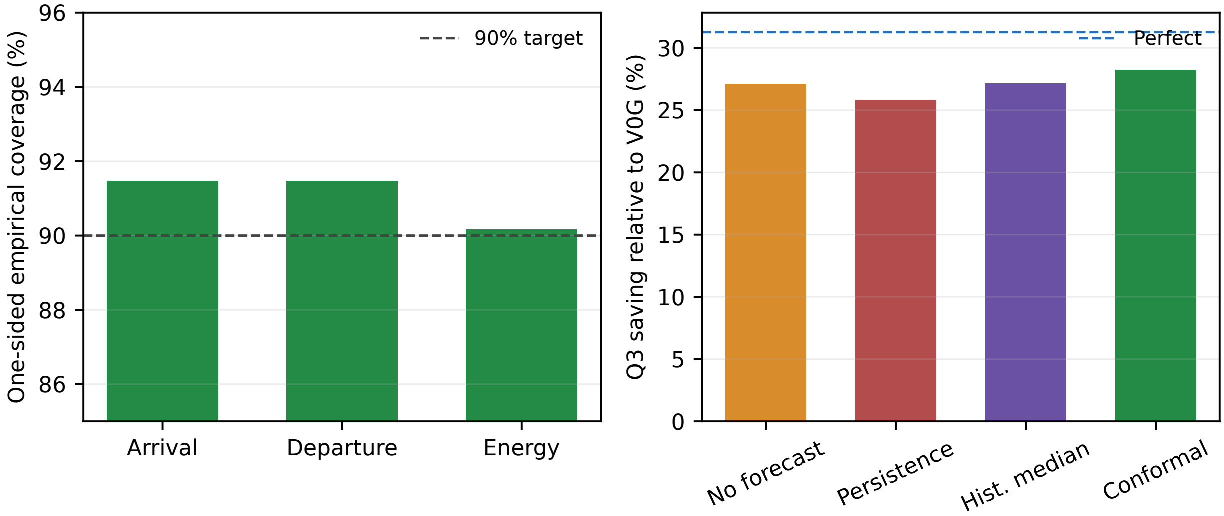

Figure 2: Left: empirical one-sided Q3 coverage of conformal corrections. Right: Q3 saving relative to V0G; the dashed line is perfect information.

Table 3: Chronological June validation used to select α. June is never part of the Q3 test.

α	Nominal coverage	June validation cost (USD)
0.1	90%	1603.32
0.2	80%	1628.86
0.3	70%	1636.76
0.4	60%	1737.04

Table 4: EV/charger invariance and synthetic scaling diagnostics.

Diagnostic	Value
Maximum absolute objective gap	1.14e-12
Maximum aggregate-load gap (kW)	7.64e-11
Median synthetic speedup at 8 sessions/charger	2.07×

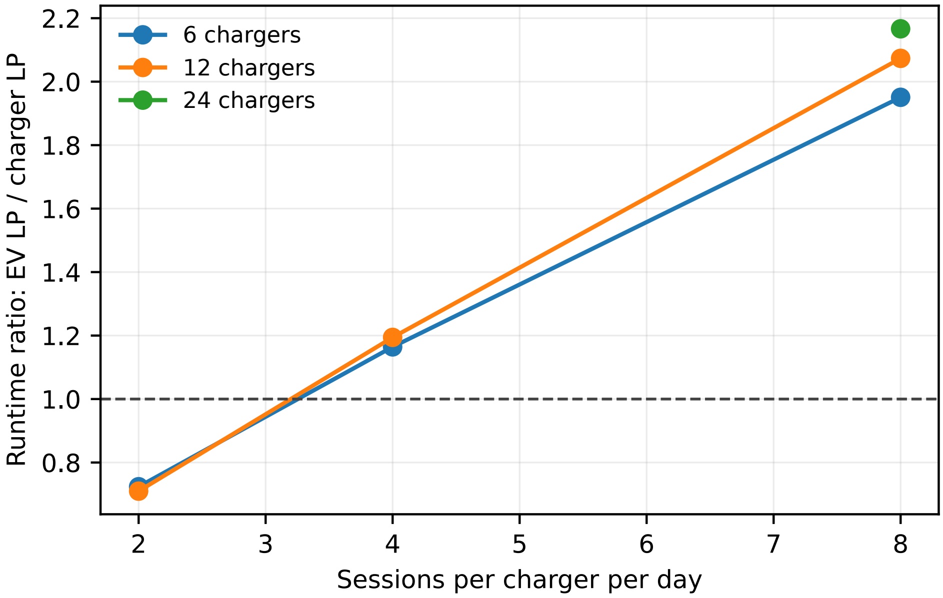

Figure 3: Median runtime ratio of the EV LP to the exact sparse charger LP. Aggregation becomes beneficial as sessions per charger increase; at low density, charger cumulative states can outweigh variable reduction.

6 Discussion and limitations

The experiment supports three conclusions. First, billing state is part of the controller state: isolated-day demand accounting is not an innocuous reporting choice. Second, charger aggregation can be exact and computationally useful, but only because physical nonoverlap enables unambiguous disaggregation. Third, a simple calibrated uncertainty correction has positive Q3 economic value, despite worse point MAE, illustrating that forecast quality should be evaluated through downstream control as well as prediction.

The evidence is deliberately bounded:

- six ports and three billing months do not establish statistical significance, seasonality robustness, or campus-wide generalization;
- the cohort's nonoverlap requirement excludes shared-port anomalies and simultaneous connector use;
- calibration is marginal and ordinal-matched; it gives neither joint trajectory coverage nor protection against missing sessions;
- the model assumes energy and departure are known at plug-in and does not model early unplugging after arrival;
- tariff coefficients reproduce the research implementation and must be reconciled with the tariff vintage used in a final submission; and
- no V2G, demand response, network power flow, PV, or stationary storage is modeled. Accordingly, "revenue" claims would be misleading; the present quantity is avoided charging cost.

A submission-strength next experiment should pre-register multiple disjoint port cohorts, extend the untouched test to at least 12 billing months, report block-bootstrap confidence intervals at the month or charger-month level, and compare learned quantile or distributional session models under the identical MPC and tariff evaluator. That design can determine whether the observed 1.47% improvement is stable rather than a Q3-specific pilot effect.

7 Conclusion

We developed a forecast-calibrated charger MPC that preserves individual session feasibility, carries monthly demand peaks correctly, and produces a fully causal Q3 comparison. In the six-port pilot, conformal robustness reduces cost relative to immediate charging and modestly improves on the causal point forecast while retaining zero unserved energy. Exact EV/charger invariance and density-dependent scaling are independently verified. The mathematical and computational foundations are paper-ready; the empirical claim remains preliminary until replicated across more cohorts and seasons.

Reproducibility statement

Source code, compact result tables, experiment commands, generated figures, and this manuscript are version controlled. The raw and processed charging tables are excluded from Git because of size and data-use restrictions. The Q3 runner uses a fixed cohort, chronological calibration, validation, and test windows, daily checkpoints, and hard failures for unserved energy, solver fallback, or invalid physical overlap.

Data availability

The charging records are held locally and their redistribution is subject to UC San Diego and source-data restrictions. Derived compact numerical results needed to reproduce all manuscript tables and figures are included with the code. Access to the underlying session table should be requested through the appropriate data owner.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported here.

Declaration of generative AI and AI-assisted technologies

During preparation of this draft, the author used an AI coding and writing assistant to help refactor code, generate tests, and edit language. All model formulations, numerical results, citations, and conclusions remain subject to the author's review and responsibility. This statement should be updated to match the journal's policy and the final author workflow before submission.

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A Implementation-to-equation audit

Traceability between the mathematical model and implementation.

Component	Equations	Implementation invariant
Session feasibility	(2)	validation, exact energy equality, and availability bounds
Charger projection	(4)–(6)	nonoverlap assertion, sparse cumulative state, and explicit disaggregation
Monthly billing	(7)–(8)	executed peak carried day to day and reset only at month boundary
Conformal correction	(10)–(12)	finite-sample order statistic and capacity clipping
Rolling information	Section 3.3	arrived truth replaces future forecast; only connected actual load is executed
Tie-breaking	(13)	primary HiGHS LP followed by a constrained secondary LP

B Artifact commands

```
PYTHONPATH=src /opt/homebrew/bin/python3 -m unittest discover -s tests -v
bash experiments/run_paper_protocol.sh
```